A fermionic normal form for the ribbon operator $R_e(t)$, and its identification with the Kashiwara–Miwa–Stern boson

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Abstract

Let $R_e(t)$ be the operator on Λ that adds a connected e-ribbon with weight t^{ht}. We prove that $R_e(t)$ is a *classical* (undeformed) free-fermion bilinear dressed by a diagonal operator: $R_e(t) = \sum_{b\in\mathbb{Z}} \psi_{b+e}\psi_b^* \, (-t)^{N_{(b,b+e)}}$, where $N_{(b,b+e)} = \sum_{b<j<b+e} n_j$ counts beads in the open interval jumped by the move. We then prove $[R_e(t)^*, R_e(t)] = 1+t^2+\cdots+t^{2(e-1)}$ by an explicit telescoping in the Maya diagram.

Verdict, stated first. The conjecture is true, and the operator is *known*: $R_e(t)$ is Thomas Lam's $B_{-1} = p_1(\mathbf{u}) = \sum_i u_i^{(e)}$ [La1], i.e. the Kashiwara–Miwa–Stern boson on the level-1 q-deformed Fock space of $U_q(\widehat{\mathfrak{sl}}_e)$, under $q = t$. The identification is *definitional*, not a shape match: Lam's defining formula for $u_i^{(n)}$ and my defining formula for $R_e(t)$ are the same formula. This is the crossing from the ribbon path into the q-Fock / integrable-lattice path, and it is a **bridge, not a defeat**. Accordingly the central term $[R_e^*, R_e] = [e]_{t^2}$ proved below is *not new* — it is Lam's Corollary (`cor:KMS`), originally Kashiwara–Miwa–Stern — though the proof given here is. What does appear to be new, and is the content of this note, is the normal form itself: the sources I hold state the operator combinatorially on partitions or inside a q-*deformed* wedge, never as an undeformed Clifford bilinear times a diagonal weight.

1 Conventions

Convention 1. Λ is the ring of symmetric functions over $\mathbb{Q}(t)$, with Schur basis $\{s_\lambda\}$ and the Hall inner product $\langle s_\lambda, s_\mu\rangle = \delta_{\lambda\mu}$.

Maya set. For a partition λ put $M(\lambda) = \{\lambda_j - j : j \geq 1\} \subset \mathbb{Z}$; it is cofinite in $\mathbb{Z}_{<0}$ and meets $\mathbb{Z}_{\geq 0}$ finitely. Elements of M are *beads*, elements of $\mathbb{Z}\setminus M$ *holes*. We write $m(x) = [x \in M]$.

Fermionic Fock space. $\mathcal{F}$ is the charge-0 semi-infinite wedge with basis $|M\rangle = v_{a_1}\wedge v_{a_2}\wedge\cdots$, where $a_1 > a_2 > \cdots$ enumerate M in *decreasing* order; we write $|\lambda\rangle := |M(\lambda)\rangle$ and identify $|\lambda\rangle \leftrightarrow s_\lambda$, so $\{|\lambda\rangle\}$ is orthonormal. The free fermions satisfy $\{\psi_a, \psi_b^*\} = \delta_{ab}$, $\{\psi_a,\psi_b\} = \{\psi_a^*,\psi_b^*\} = 0$, with $\psi_a|M\rangle = v_a \wedge |M\rangle$ (zero if $a \in M$) and ψ_a^* its adjoint. *The Clifford algebra is undeformed: no parameter t or q occurs in these relations.* Set $n_a = \psi_a\psi_a^*$, so $n_a|M\rangle = m(a)\,|M\rangle$.

Ribbon operator. For $e \geq 1$,

$$R_e(t)\, s_\lambda \;=\; \sum_\mu t^{\mathrm{ht}(\mu/\lambda)}\, s_\mu,$$

the sum over $\mu \supset \lambda$ with μ/λ a connected border strip (=ribbon) of size e, and $\mathrm{ht}(\mu/\lambda) =$ (number of rows of μ/λ) $- 1$.

Remark 2. ht is Lam's spin: [La1, §2] defines "*the spin* $\mathrm{spin}(R)$ *of a ribbon R is equal to the number of rows in the ribbon, minus 1*". Some authors halve it; we do not.

2 The abacus dictionary

Lemma 3 (hook lengths from the Maya set)**.** *Let λ be a partition with Maya set M. The cells of λ are in bijection with pairs (i,k) with $i \notin M$, $k \in M$, $i < k$; the cell attached to (i,k) lies in the row indexed by the bead k and has hook length $k-i$.*

Proof. Row j of λ corresponds to the bead $k = \lambda_j - j$, and we claim $\lambda_j = \#\{i < k : \ i \notin M\}$. Count in the window $[-L, k)$ for L large. It contains $k+L$ integers. Its beads are exactly $\lambda_r - r$ for $r > j$ with $\lambda_r - r \geq -L$; for $r > \ell(\lambda)$ this reads $-r \geq -L$, so r ranges over $j+1, \dots, L$ and there are $L-j$ of them. Hence the holes below k number $(k+L)-(L-j) = k+j = \lambda_j$, independently of L. Attaching to the c-th hole below k (counted downwards) the c-th cell of row j gives the bijection. That the associated quantity $k-i$ is the hook length of that cell is the classical β-set description of hook lengths (Macdonald I.1, Ex. 8): the hook lengths of row j are precisely $\{k-i : \ i<k, \ i \notin M\}$. ☐

Lemma 4 (ribbons are bead moves; height counts jumped beads)**.** *Let λ have Maya set M and let $e \geq 1$. The assignment $b \mapsto \mu$ with $M(\mu) = \big(M \setminus \{b\}\big) \cup \{b+e\}$ is a bijection*

$$\{\, b \in \mathbb{Z} : \ b \in M, \ b+e \notin M \,\} \ \xrightarrow{\ \sim\ } \ \{\, \mu \supset \lambda : \ \mu/\lambda \textit{ a connected } e\textit{-ribbon} \,\},$$

and

$$\operatorname{ht}(\mu/\lambda) \ = \ \#\big(M \cap (b, b+e)\big).$$

Moreover only finitely many b occur.

Proof. By Lemma 3 a hook of length e in μ is a pair $(b, b+e)$ with $b+e \in M(\mu)$, $b \notin M(\mu)$; removing the corresponding border strip is the classical statement that it replaces the bead $b+e$ by b (James–Kerber 2.7.13). We verify the two quantitative claims directly.

Write $B = \#\big(M \cap (b, b+e)\big)$ and $H = \#\big((b,b+e) \setminus M\big)$, so $B + H = e-1$. Compare row lengths of λ (beads $k' \in M$) and μ (beads $k' \in M' = M \setminus \{b\} \cup \{b+e\}$); by Lemma 3 the row attached to a bead k' has length $\#\{i < k' : i \notin (\text{the Maya set})\}$.

- If $k' \in M$ with $k' > b+e$: both changes lie below k' — b becomes a hole ($+1$ hole) and $b+e$ becomes a bead (-1 hole) — so they cancel and the row is unchanged. If $k' < b$: neither change lies below k', so the row is unchanged.
- If $k' \in M \cap (b, b+e)$: the site b lies below k' and leaves M, so it becomes one extra hole below k'; the site $b+e$ lies above k' and is irrelevant. So M' has exactly one more hole below k' than M, and the row is one cell longer in μ. There are B such rows, each gaining exactly 1.
- The moving bead: in λ its row (bead b) has $\#\{i < b : i \notin M\}$ cells; in μ its row (bead $b+e$) has $\#\{i < b+e : i \notin M'\}$ cells. The difference is $\#\{i \in [b, b+e) : i \notin M\} = 1 + H$ (the site b is a hole of M', plus the H interior holes).

Total gain $= B + (1+H) = B+H+1 = e$, confirming $|\mu/\lambda| = e$. The rows that change are exactly those attached to beads of M' in $(b, b+e]$, namely the B interior beads together with the moved bead; since the beads are read in decreasing order these are $B+1$ *consecutive* rows. Hence μ/λ occupies $B+1$ rows and $\operatorname{ht}(\mu/\lambda) = B$, as claimed.

Finiteness: $M \supseteq \mathbb{Z}_{\leq c}$ for some c, so $b \leq c - e$ forces $b+e \in M$; and $b \in M$ forces b bounded above. ☐

3 The fermionic sign, and the dressing

Lemma 5 (the sign is the height sign)**.** *Let $b \in M$, $b+e \notin M$ and $M' = M \setminus \{b\} \cup \{b+e\}$. Then*

$$\psi_{b+e}\psi_b^* \,|M\rangle \;=\; (-1)^{\#(M\cap(b,b+e))}\,|M'\rangle.$$

If $b \notin M$ or $b+e \in M$ then $\psi_{b+e}\psi_b^*|M\rangle = 0$.

Proof. Let $p = \#\{a \in M : a > b\}$, so b is the $(p+1)$-st entry of the decreasing wedge. Moving v_b to the front costs $(-1)^p$, and contracting gives $\psi_b^*|M\rangle = (-1)^p\,|M \setminus \{b\}\rangle$. Let $q = \#\{a \in M \setminus \{b\} : a > b+e\} = \#\{a \in M : a > b+e\}$ (as $b < b+e$). Prepending v_{b+e} and sorting it past those q larger entries costs $(-1)^q$: $\psi_{b+e}|M \setminus \{b\}\rangle = (-1)^q|M'\rangle$. Hence the sign is $(-1)^{p+q} = (-1)^{p-q}$ and

$$p - q = \#\{a \in M : b < a \le b+e\} = \#\big(M \cap (b, b+e)\big),$$

the last step because $b+e \notin M$. The vanishing statements are immediate from $\psi_b^*|M\rangle = 0$ for $b \notin M$ and $\psi_{b+e}|M''\rangle = 0$ for $b+e \in M''$. □

Lemma 6 (the dressing is well defined)**.** *Fix b and $e \ge 1$, and set $N_{(b,b+e)} := \sum_{b<j<b+e} n_j$. Then*

$$\prod_{j=b+1}^{b+e-1} \big(1 + (-t-1)\,n_j\big) \;=\; (-t)^{N_{(b,b+e)}},$$

and this operator commutes with $\psi_{b+e}\psi_b^*$ and with $\psi_b\psi_{b+e}^*$. Consequently the dressing may be applied before or after the bead move, with the same result.

Proof. Each n_j is idempotent with eigenvalues $0, 1$; $1+(-t-1)\cdot 0 = 1$ and $1+(-t-1)\cdot 1 = -t$, so the factor at j contributes $(-t)^{n_j}$, and the factors commute. For the commutation, $\psi_{b+e}\psi_b^* = E_{b+e,b}$ and $n_j = E_{jj}$ in the $\mathfrak{gl}_\infty$ notation $E_{xy} = \psi_x\psi_y^*$, and $[E_{xy}, E_{zw}] = \delta_{yz}E_{xw} - \delta_{wx}E_{zy}$; with $\{x, y\} = \{b+e, b\}$ and $z = w = j \in (b, b+e)$ both Kronecker deltas vanish. (Computationally: over 2386 bead moves with $e \le 6$, $|\lambda| \le 9$, the count $\#(M \cap (b, b+e))$ was equal before and after the move in every case.) □

4 The normal form

Theorem 7 (fermionic normal form)**.** *For every $e \ge 1$, as operators on $\mathcal{F} \cong \Lambda$,*

$$\boxed{R_e(t) \;=\; \sum_{b\in\mathbb{Z}} \;\psi_{b+e}\,\psi_b^* \prod_{j=b+1}^{b+e-1} \big(1 + (-t-1)\,n_j\big) \;=\; \sum_{b\in\mathbb{Z}} \;\psi_{b+e}\,\psi_b^*\,(-t)^{N_{(b,b+e)}}}$$

the sum being locally finite.

Proof. Apply the right-hand side to $|M\rangle = |\lambda\rangle$. By Lemma 6 the b-th summand is $(-t)^{\#(M\cap(b,b+e))}\,\psi_{b+e}\psi_b^*|M\rangle$, which by Lemma 5 vanishes unless $b \in M$ and $b+e \notin M$, and otherwise equals

$$(-1)^{\#(M\cap(b,b+e))}(-t)^{\#(M\cap(b,b+e))}\,|M'\rangle \;=\; t^{\#(M\cap(b,b+e))}\,|M'\rangle.$$

By Lemma 4 the surviving b are finite in number and biject with the μ such that μ/λ is a connected e-ribbon, with $\#(M \cap (b, b+e)) = \mathrm{ht}(\mu/\lambda)$. Summing gives $\sum_\mu t^{\mathrm{ht}(\mu/\lambda)}|\mu\rangle = R_e(t)|\lambda\rangle$. □

Corollary 8 (the anchor is the vanishing of one scalar)**.** *At $t=-1$ the scalar $(-t-1)$ vanishes, every factor of the dressing becomes* 1*, and*

$$R_e(-1) \;=\; \sum_{b\in\mathbb{Z}} \psi_{b+e}\psi_b^* \;=\; \alpha_{-e} \;=\; M_{p_e},$$

the Murnaghan–Nakayama rule. Thus the discontinuity of $\operatorname{ord} R_e(c)$ *at $c=-1$ (zero there, infinite at every other scalar) is exactly the vanishing of the single scalar $1+t$: it is the point at which the diagonal dressing is invisible.*

Proof. The first equality is the substitution $t=-1$ in Theorem 7. $\sum_b \psi_{b+e}\psi_b^*$ is the Heisenberg mode α_{-e} (no normal-ordering correction is needed for $e\geq 1$), which under the boson–fermion correspondence is multiplication by p_e; equivalently, the case $t=-1$ of Theorem 7 *is* Murnaghan–Nakayama. Independently verified: $R_e(-1)s_\lambda = p_e\, s_\lambda$ on 90/90 pairs ($e\leq 4, |\lambda|\leq 6$) with $p_e\, s_\lambda$ computed in the monomial basis via Kostka numbers, an engine that uses neither Murnaghan–Nakayama nor the abacus. ☐

5 The central term, and the separator

The following is the test I used to decide whether $R_e(t)$ is a known operator. It is a coordinate with no free parameter to fit.

Theorem 9. *Let R_e^* denote the adjoint of $R_e(t)$ for the Hall inner product, i.e. $R_e^* s_\mu = \sum_\lambda t^{\mathrm{ht}(\mu/\lambda)} s_\lambda$ over connected e-ribbons μ/λ. Then*

$$\big[\,R_e(t)^*,\, R_e(t)\,\big] \;=\; \big(1+t^2+t^4+\cdots+t^{2(e-1)}\big)\cdot \mathrm{Id} \;=\; [e]_{t^2}\cdot \mathrm{Id}.$$

Proof. Write $A_b=\psi_{b+e}\psi_b^* = E_{b+e,b}$, $A_b^\dagger = \psi_b\psi_{b+e}^* = E_{b,b+e}$ and $D_b = (-t)^{N_{(b,b+e)}}$. By Theorem 7, $R_e = \sum_b A_b D_b$; since the D_b are diagonal with t-coefficients and $\{|\lambda\rangle\}$ is orthonormal, $R_e^* = \sum_b D_b A_b^\dagger = \sum_b A_b^\dagger D_b$ (Lemma 6).

Step 1: the off-diagonal terms vanish. Fix $b\neq c$ and set $\alpha = [\,b<c<b+e\,]$, $\beta=[\,c<b<c+e\,]$ (not both 1). Since A_c moves a bead from c to $c+e$, on its support the count $N_{(b,b+e)}$ changes by $-\alpha+\beta$ — indeed c leaves the open interval $(b,b+e)$ iff $\alpha=1$, and $c+e$ enters it iff $b<c+e<b+e$, i.e. iff $c<b<c+e$, i.e. $\beta=1$. Hence $D_bA_c = (-t)^{\beta-\alpha}A_cD_b$ as operators. Symmetrically $A_b^\dagger$ moves a bead from $b+e$ to b and changes $N_{(c,c+e)}$ by $-[c<b+e<c+e]+[c<b<c+e] = -\alpha+\beta$, so $D_cA_b^\dagger = (-t)^{\beta-\alpha}A_b^\dagger D_c$. Finally $[A_b^\dagger, A_c] = [E_{b,b+e}, E_{c+e,c}] = \delta_{b+e,c+e}E_{b,c} - \delta_{c,b}E_{c+e,b+e} = 0$ for $b\neq c$. Therefore

$$A_b^\dagger D_b A_c D_c - A_c D_c A_b^\dagger D_b = (-t)^{\beta-\alpha}\big(A_b^\dagger A_c - A_c A_b^\dagger\big) D_b D_c = 0.$$

Step 2: the diagonal terms. For $c=b$, D_b commutes with both A_b and $A_b^\dagger$, so the term is $[A_b^\dagger, A_b]D_b^2 = (E_{bb}-E_{b+e,b+e})\,t^{2N_{(b,b+e)}} = (n_b - n_{b+e})\,t^{2N_{(b,b+e)}}$. Hence

$$[R_e^*, R_e] \;=\; \sum_{b\in\mathbb{Z}} (n_b - n_{b+e})\, t^{2N_{(b,b+e)}},$$

which is diagonal in the basis $\{|M\rangle\}$ with eigenvalue

$$S(M) \;=\; \sum_{b\in\mathbb{Z}} f(b), \qquad f(b) := \big(m(b)-m(b+e)\big)\,t^{2N_b}, \quad N_b := \#\big(M\cap(b,b+e)\big).$$

(The sum is finite: for $b \ll 0$ both $m(b) = m(b+e) = 1$, for $b \gg 0$ both vanish.)

Step 3: the telescoping. Define the potential

$$g(b) \;:=\; \sum_{i=0}^{e-1} m(b+i)\, t^{2\,c_i(b)}, \qquad c_i(b) := \#\big(M \cap (b+i,\, b+e)\big).$$

We claim $f(b) = g(b) - g(b+1)$ for every b. Put $X := \sum_{i=1}^{e-1} m(b+i)t^{2c_i(b)}$ and $\epsilon := m(b+e)$, and note $c_0(b) = N_b$, so $g(b) = m(b)t^{2N_b} + X$. For $1 \le i' \le e-1$, $c_{i'}(b+1) = \#(M \cap (b+i', b+e+1)) = c_{i'}(b) + \epsilon$, while the term $i' = e$ of $g(b+1)$ has empty interval and weight t^0. Hence

$$g(b+1) = t^{2\epsilon} X + \epsilon.$$

If $\epsilon = 0$: $g(b) - g(b+1) = m(b)t^{2N_b} + X - X = m(b)t^{2N_b} = f(b)$. If $\epsilon = 1$: let the beads of M inside $(b, b+e)$ sit at $b + i_1 < \cdots < b + i_{N_b}$; then $c_{i_k}(b) = N_b - k$, so $X = \sum_{k=1}^{N_b} t^{2(N_b-k)} = [N_b]_{t^2}$ and $(1-t^2)X = 1 - t^{2N_b}$. Thus

$$g(b) - g(b+1) = m(b)t^{2N_b} + (1-t^2)X - 1 = m(b)t^{2N_b} - t^{2N_b} = \big(m(b) - 1\big)t^{2N_b} = f(b).$$

Finally $g(b) \to 0$ as $b \to +\infty$ (all $m = 0$), and for $b \ll 0$ all $m = 1$ and $c_i(b) = e - 1 - i$, so $g(b) = \sum_{i=0}^{e-1} t^{2(e-1-i)} = [e]_{t^2}$. Therefore $S(M) = \sum_b \big(g(b) - g(b+1)\big) = [e]_{t^2}$, independently of M. □

Remark 10 (what Theorem 9 is, and is not)**.** Theorem 9 is **not new**. It is [La1, Cor. `cor:KMS`, eq. (`hei`)] specialised to $k = 1$: Lam states $[B_k, B_l] = k\, \frac{1-q^{2n|k|}}{1-q^{2|k|}}\, \delta_{k,-l}$ and attributes it to Kashiwara–Miwa–Stern [KMS]; at $k = 1$, $n = e$, $q = t$ this is $1 + t^2 + \cdots + t^{2(e-1)}$. Uglov's γ_m at $l = 1$ [Ug, Prop. `p:gamma`] gives the same in the inverse convention, $\gamma_1 = \frac{1-q^{-2n}}{1-q^{-2}} = [n]_{q^{-2}}$. What is new here is the proof — a two-line Clifford computation plus an explicit telescoping potential g — and, more importantly, the *use*: it was the untuned coordinate that identified the operator (§8). Its consistency with Corollary 8 is a further check: $[e]_{t^2}\big|_{t=-1} = e$, matching $[\,p_e^{\perp}, M_{p_e}\,] = e$.

Corollary 11. *For $e \ne f$ the operators $R_e(t), R_f(t)$ do* not *commute, but $(1+t)$ divides every coefficient of $[R_e(t), R_f(t)]$.*

Proof. Non-commutation is exhibited already on $s_\emptyset$: $[R_1, R_2]s_\emptyset = (1+t)\, s_{(2,1)}$. Divisibility: at $t = -1$, Corollary 8 gives $R_e(-1) = M_{p_e}$ and $R_f(-1) = M_{p_f}$, which commute as multiplication operators; hence every coefficient of $[R_e, R_f]$, a polynomial in t, vanishes at $t = -1$. (Verified: 486 commutator coefficients over $(e, f) \in \{(1,2), (1,3), (2,3), (2,4), (3,4), (3,5)\}$, $|\lambda| \le 5$; all vanish at $t = -1$.) This is not in tension with the Heisenberg structure: in the Kashiwara–Miwa–Stern picture R_e and R_f are the B_{-1}'s of *different* ranks $n = e$ and $n = f$, acting on the same space but belonging to different Heisenberg subalgebras. □

6 Occupation numbers have infinite differential order

Recall Grothendieck's order filtration on operators on the commutative ring Λ: $\mathrm{ord}(D) \le d$ iff every $(d+1)$-fold nested bracket $[\cdots[[D, M_{f_1}], M_{f_2}], \ldots, M_{f_{d+1}}]$ vanishes; since Λ is generated by the p_a it suffices to take $f_i \in \{p_a\}$. One has $\mathrm{ord}(M_f) = 0$ and $\mathrm{ord}(p_\rho^{\perp}) = \ell(\rho)$.

Proposition 12. *For every $j \in \mathbb{Z}$, $\mathrm{ord}(n_j) = \infty$.*

Proof. Let $A = M_{p_1}$, whose matrix entries in the Schur basis are $A_{\mu\lambda} = [\,\mu/\lambda \text{ is one box}\,]$ (Pieri). For a diagonal operator D with $Ds_\lambda = \nu(\lambda)s_\lambda$ one checks by induction that

$$\left(\mathrm{ad}_A^n D\right)_{\mu\lambda} \;=\; \sum_{\lambda=\lambda_0\subset\lambda_1\subset\cdots\subset\lambda_n=\mu} \;\sum_{i=0}^{n}(-1)^{n-i}\binom{n}{i}\,\nu(\lambda_i),$$

the outer sum over chains adding one box at a time. Take $D = n_j$, so $\nu(\lambda) = m_\lambda(j) := [\,j \in M(\lambda)\,]$. We exhibit, for each $n \geq 1$, a pair (λ, μ) joined by a *unique* chain and with nonzero inner sum.

Case $j \geq 0$. Take $\lambda = (j{+}1)$, $\mu = (j{+}1{+}n)$. The only partitions between them are the one-row shapes $(j+1+i)$, so the chain is unique. Now $M((r)) = \{r-1\} \cup \mathbb{Z}_{\leq -2}$, so for $j \geq 0$ we have $m_{(r)}(j) = [\,r = j+1\,]$; along the chain $\nu(\lambda_i) = [\,i = 0\,]$ and the inner sum is $(-1)^n\binom{n}{0} = (-1)^n \neq 0$.

Case $j = -1$. Take $\lambda = \emptyset$, $\mu = (n)$; again a unique chain, and $m_{(r)}(-1) = [\,r = 0\,]$, so $\nu(\lambda_i) = [\,i = 0\,]$ and the sum is $(-1)^n \neq 0$.

Case $j \leq -2$. Take $\lambda = (1^{-j})$, $\mu = (1^{-j+n})$; the only partitions between two columns are columns, so the chain is unique. Here $M((1^m)) = \mathbb{Z}_{\leq 0} \setminus \{-m\}$, so $m_{(1^m)}(j) = [\,j \neq -m\,]$ and along the chain $\nu(\lambda_i) = 1 - [\,i = 0\,]$. Since $\sum_i (-1)^{n-i}\binom{n}{i} = 0$ for $n \geq 1$, the inner sum is $-(-1)^n \neq 0$.

In every case $\mathrm{ad}^n_{M_{p_1}}(n_j) \neq 0$ for all n, so $\mathrm{ord}(n_j) = \infty$. □

Remark 13 (what this does and does not explain)**.** Theorem 7 writes $R_e(t)$ as a *finite* expression in the undeformed Clifford algebra — fermionic degree at most $2e$ — whose only t-dependence is diagonal. Proposition 12 says the diagonal factors are individually of infinite differential order. Together these *situate* the result $\mathrm{ord}\,R_e(t) = \infty$ for $t \neq -1$: the infinitude is a statement about the bosonic coordinates $p_1, p_2, \ldots$, not about the operator's complexity on the bead side.

It must be said plainly that this is an explanation and not a proof of that result. Infinite sums of infinite-order operators can have finite order: the Euler operator $E = \sum_{n\geq 1} M_{p_n}p_n^\perp$ has $\mathrm{ord}(E) = 1$ and $Es_\lambda = |\lambda|s_\lambda$, yet E is the regularised $\sum_j j\,n_j$ (because $\sum_i(\lambda_i - i) - \sum_i(-i) = |\lambda|$). So no argument of the form "R_e is built from n_j's, each of infinite order, hence R_e has infinite order" is valid. The infinite order of $R_e(t)$ for $t \neq -1$ rests on its own proof.

7 Computational verification

All checks use two *code-disjoint* engines. The *shape* engine (`probes/2026-09-06-Q84/engine.py`) enumerates skew shapes with one edge-connected component and no 2×2 square and weights them t^{ht}. The *bead* engine (`probes/2026-09-06-c2-Q91/bead.py`) was written fresh for this note; it imports nothing from the shape engine and computes the sign of $\psi_{b+e}\psi_b^*$ by literally sorting the wedge — so Lemma 5 is *tested*, not assumed. t is a sympy symbol throughout; agreement means equality of polynomials in t.

check	range	result
Theorem 7, shape = bead	$e \in \{2,3,4\}$, $\lvert\lambda\rvert \leq 7$	135/135
Theorem 7, wider	$e \in \{2,\ldots,6\}$, $\lvert\lambda\rvert \leq 9$	485/485
Lemma 6 (before = after)	2386 bead moves	0 disagreements
Cor. 8, $R_e(-1) = M_{p_e}$, third engine	$e \leq 4$, $\lvert\lambda\rvert \leq 6$	90/90
Thm. 9, $[R_e^*, R_e] = [e]_{t^2}$	$e \leq 6$, $\lvert\lambda\rvert \leq 7$	45/45 each e
Thm. 9, telescoping $f = g(b) - g(b{+}1)$	$e \leq 5$, $\lvert\lambda\rvert \leq 7$	0 failing b
Prop. 12, predicted witness sign	$j \in [-3, 2]$, $n \leq 5$	exact match

The named non-degenerate witness

The conjecture is vacuous where the dressing has at most one nontrivial factor. The smallest instance with $\#(M \cap (b, b+e)) \geq 2$ — so that the product $\prod(1 + (-t-1)n_j)$ genuinely has two nontrivial factors — is

$$\boxed{\lambda = \emptyset, \quad e = 3, \quad b = -3}$$

Here $M(\emptyset) = \mathbb{Z}_{<0}$, the open interval $(-3, 0)$ contains the two beads $-2, -1$, both factors equal $-t$, the fermionic sign is $(-1)^2 = +1$, and the coefficient of $s_{(1,1,1)}$ is t^2. Over the tested range the open-interval occupancy reaches $e-1$ for every $e \in \{2, \ldots, 6\}$, i.e. up to 5; 163 coefficients of t-degree ≥ 2 occur.

Planted-defect controls, and the kernel of the harness

Ten variants were run against the shape engine on 135 data points. Five *distinct* defects were observed to fail:

variant		agreement	
V0	conjecture: sign $\cdot\, (-t)^{\#(M\cap(b,b+e))}$	135/135	PASS
V2	closed interval $[b, b+e]$	0/135	FAIL
V3	exponent $N+1$	0/135	FAIL
V4	t^N instead of $(-t)^N$	0/135	FAIL
V5	wrong window $(b-e, b)$	12/135	FAIL
V6	holes: exponent $(e-1)-N$	0/135	FAIL
V7	fermionic sign dropped, $(-t)^N$	0/135	FAIL
V9	dressing dropped (bare bilinear)	0/135	FAIL

Testing the variants *against each other* showed the harness has a kernel, which must be recorded because it means some "controls" are not controls:

$$\text{V0} \equiv \text{V1} \equiv \text{V8}, \qquad \text{V2} \equiv \text{V3}, \qquad \text{V4} \equiv \text{V7} \quad \text{(identically, on all 135 points)}.$$

The reasons are structural, not numerical. **(i)** A valid move forces $b \in M$ and $b+e \notin M$, so *both endpoints of the interval carry no information*: replacing $(b, b+e)$ by $(b, b+e]$ changes nothing (V1≡V0), and replacing it by $[b, b+e]$ is exactly the shift $N \mapsto N+1$ (V2≡V3). The two controls prescribed as "closed interval" and "exponent $N+1$" are therefore *one* control. **(ii)** V0≡V8 ("sign dropped, t^N") is precisely the content of Lemma 5, sign $= (-1)^N$; that V8 passes while V7 fails is thus an independent confirmation of the sign lemma rather than a defective control. Because t is symbolic, comparing degrees forces $N = \mathrm{ht}$ and comparing coefficients then forces sign $= (-1)^{\mathrm{ht}}$: the two halves of the theorem are separated by the check. At a numerical t they would not be — at $t = -1$ the dressing is blind, at $t = 1$ the exponent is.

8 Literature: this operator is known

The identification. Lam [La1, §2] defines, for fixed n, the *ribbon Schur operators*

$$u_i^{(n)} : \lambda \longmapsto \begin{cases} q^{\mathrm{spin}(\mu/\lambda)}\,\mu & \text{if } \mu/\lambda \text{ is an } n\text{-ribbon with head on the } i\text{-th diagonal,} \\ 0 & \text{otherwise,} \end{cases}$$

with spin = (rows) − 1, and their adjoints d_i for the inner product making $\{\lambda\}$ orthonormal. Summing over all diagonals,

$$\sum_{i\in\mathbb{Z}} u_i^{(e)}\Big|_{q=t} \;=\; R_e(t), \qquad \sum_{i\in\mathbb{Z}} d_i^{(e)}\Big|_{q=t} \;=\; R_e(t)^*.$$

This is a *definitional* identity: the two defining formulas are the same formula, the head-diagonal index merely partitioning the sum. By [La1, Cor. `cor:KMS`], $B_{-k} = p_k(\mathbf{u})$ and $B_k = p_k^\perp(\mathbf{u})$ generate the Kashiwara–Miwa–Stern [KMS] Heisenberg action on the level-1 q-deformed Fock space of $U_q(\widehat{\mathfrak{sl}}_n)$. Since $p_1 = e_1 = h_1$, we get $B_{-1} = \sum_i u_i$, so

$$\boxed{R_e(t) \;=\; B_{-1} \;\text{ at rank } n = e,\; q = t.}$$

Theorem 9 is then Lam's eq. (`hei`) at $k = 1$, and the agreement of $[e]_{t^2}$ with $[n]_{q^2}$ was the untuned coordinate that confirmed the identification. (Related normalisations: Lam [La2, `prop:hor`] gives $V_k|\lambda\rangle = \sum(-q)^{-s(\mu/\lambda)}|\mu\rangle$ over horizontal n-ribbon strips of size k, whose $k = 1$ case is the same operator at $t = -q^{-1}$; my own earlier file `proofs/2026-08-14-C4-iijima-B1.tex` derived $B_{-1}|\lambda\rangle = \sum(-q^{-1})^h|\mu\rangle$ from Uglov Prop. 3.16, i.e. the same operator in that convention.)

This is the crossing, not a defeat. The identification is exactly the bridge the ribbon programme has been looking for: $R_e(t)$, reached from Murnaghan–Nakayama and border-strip combinatorics, is a level-1 q-Fock-space object, which places the whole family inside LLT/KMS/Uglov territory where $U_q(\widehat{\mathfrak{sl}}_e)$, canonical bases and the vertex-model machinery are available.

What appears to be new. Neither Lam [La1, La2], nor KMS [KMS], nor Uglov [Ug] writes this operator in the undeformed Clifford algebra. Lam works combinatorially on partitions — his two papers contain no occurrence of *fermion*, *Clifford*, ψ, *wedge*, *occupation*, *Maya*, *abacus* or *bead* (checked by grep at source; the single hit in [La2] is a bibliography entry for Hayashi). KMS and Uglov work inside a *q-deformed* wedge with straightening rules; Uglov's B_m (`e:Bo`) is given on wedges and never evaluated in ribbon language (*ribbon*, *spin*: 0 hits in `cbf4.tex`). Theorem 7 says something different from all of these: the q-deformation can be moved entirely out of the algebra and into a *diagonal* weight over the *undeformed* Clifford algebra.

Honest limits of that novelty claim. This is an absence measured over the sources on my disk, which is the weak kind of claim. A normal form of this shape is the sort of thing that could sit unremarked in a physics paper on $W_{1+\infty}$ or in the Bloch–Okounkov literature under the name of a diagonal "energy" dressing rather than a ribbon height; no keyword crosses between "spin", "height" and "occupation number". I have *not* read Bloch–Okounkov, and [La1] was recorded in my own index at `agent-summary` level while its full LaTeX source sat on disk unread — an index defect corrected in the course of this note. The identification in the first paragraph is solid; the novelty claim in this one is not, and should not be repeated without a literature search aimed at $W_{1+\infty}$.

9 Gaps and what was not done

1. **Lemma 3** cites Macdonald I.1 Ex. 8 for the identification of $\{k - i : i < k,\; i \notin M\}$ with the hook lengths of row k; I did not reprove it. Lemma 4's quantitative content (the e-cell count and the height) *is* proved here and independently verified.

2. **Connectivity** of the border strip in Lemma 4 is quoted from James–Kerber 2.7.13; what I prove directly is that the affected rows are $B+1$ consecutive rows gaining e cells in total. A fully self-contained proof would also check the interlacing that makes the union a ribbon.

3. **Proposition 12** does not imply $\operatorname{ord} R_e(t) = \infty$; see the remark following it. That statement rests on its own proof (`proofs/2026-09-06-Q84-order-of-N-e.tex`).

4. **Q86's flux** (a Boltzmann weight on bead gaps, testing $\Phi_W + \Phi_S = \Phi_E + \Phi_N$) was not attempted; the identification with B_{-1} makes it a question about the KMS Heisenberg rather than a new construction, which changes how it should be posed.

5. **The novelty of Theorem 7** is unresolved pending a $W_{1+\infty}$/Bloch–Okounkov search, which this session's rules forbade.

References

[KMS] M. Kashiwara, T. Miwa, E. Stern, *Decomposition of q-deformed Fock spaces*, `q-alg/9508006`.

[La1] T. Lam, *Ribbon Schur operators*, `math/0409463`; §2 (definition of $u_i^{(n)}$, spin), §(Cauchy) (adjoints d_i), Cor. `cor:KMS` eq. (`hei`).

[La2] T. Lam, *Ribbon tableaux and the Heisenberg algebra*, `math/0310250`; Prop. `prop:hor`, Prop. `prop:bkaction`.

[Ug] D. Uglov, *Canonical bases of higher-level q-deformed Fock spaces and Kazhdan–Lusztig polynomials*, `math/9905196`; eq. (`e:Bo`), Prop. `p:gamma`, Prop. 3.16.